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J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai



Department of CSE

"A place to learn; A chance to grow"
2025-2026(Even Semester)

MSE-2 EXAMINATION

Branch: CSE A & B

Subject: Augmented Reality (AR)

Max Marks: 20

Year/Semester: 6th Semester

Subject Code: CS6E005

Duration: - 1 Hr.

Date: 06/04/2026

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO1 To understand the basic concept and framework of virtual reality

CO2 To categories the technology for multimodal user interaction and perception in VR.

CO3 To evaluate algorithmic strategies to solve a given problem.

CO4 To apply VR Tools in real time environment.

CO5 To implement application of AR/VR technology with hands on experience through more informative and practical exploration.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q.

No.

Questions

Level/CO
Mapped
L2/CO3

Marks

Q.1 A VR scene contains 6 objects with the following Z coordinates:

1

Object	Z Position
A	3
B	5
C	9
D	13
E	-3
F	7

Camera frustum limits:

Near plane = -1

Far plane = 11

Identify which objects will be rendered.

Q.2 Explain occlusion.

L2/CO3

1

Q.3 A virtual object of mass 2kg is pushed with a force of 10N

L5/CO4

1

Evaluate the acceleration.

Q.4 State the working of a physics engine.

L1/CO4

1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence).

Q.5 Define Tessellation and does it increase or decrease polygons. Explain with suitable examples.

L1/CO3

2

Q.6 State the 3 components of FBM model.

L1/CO4

2

Q.7 Discuss Unity game engine.

L2/CO5

2

Q.8 A VR city scene contains 20 buildings.

L4/CO3

2

A large building blocks 8 buildings behind it from the camera view.

Calculate how many buildings will be rendered.

SECTION C

1. Solve any two questions.

Q.9 Explain the difference between local and world coordinates with examples?

L2/CO3

4

Q.10 A virtual spring has:

$K = 300 \text{ N/m}$

$x = 0.02 \text{ m}$

Formulate the stored energy?

L5/CO4

4

Q.11 Given:

Ambient intensity $I_a = 0.3$

Ambient coefficient $k_a = 0.5$

Light intensity $I = 1.0$

Diffuse coefficient $k_d = 0.4$

Angle $\theta = 90^\circ$

Specular coefficient $k_s = 0.3$

Angle $\alpha = 0^\circ$

Shininess $n = 2$

Predict the total phong intensity.

L5/CO3

4

20 - 8

12